

Week	Lecture (Tue, 18:30pm) – LT16	Deadlines
1	L0: Course Admin L1: Introduction	
2	L2: Methodologies and Capacity Planning Lstats (only lecture notes): Review of Probability and Statistics Concepts	
3	L3: Introduction to Queueing Theory	
4	L4: Markovian Queueing Models	
5	L5: Operational Analysis	
6	L6: Background in OS Test 1 (open-book, in-person)	Tue, 18:30 – Test 1
Recess		
7	L7: Observability Tools	
8	L8: Resources Analysis (System)	
9	L9: Monitoring and Visualization	
10	L10: Performance analysis on GPGPU	
11	L11: Applications and Benchmarking L12: Case studies	Tue, 18:30 – Paper review due
12	Test 2 (open-book, in-person)	Tue, 18:30 - Test 2
13	Student Presentations	Tue, 18:30 - Presentations
Reading	No exam	Tue, 18:30 - Project due

CS5239 Performance Analysis of Computer Systems

Course coordinator: Dr. Cristina Carburaru (cristina@nus.edu.sg)

CS5239 introduces students to analytical and measurement techniques for understanding and improving computer systems performance. The course emphasizes analytical modelling approaches, such as operational analysis and stochastic queueing network models, which provide a framework for reasoning about bottlenecks, throughput limits, and end-to-end latency in complex systems. Building on this foundation, the course transitions into practical measurement methodologies such as monitoring, instrumentation, profiling, workload characterization, and benchmarking. Students learn to collect performance data using observability tools, interpret metrics, validate hypotheses, and diagnose performance issues across hardware and software layers. By the end of the course, students gain a working understanding of performance modelling, measurement, and bottleneck analysis, and are prepared to analyse and reason about system performance in a systematic manner.

After the course, students should be able to:

- Explain and interpret key performance metrics, including latency, utilization, and saturation.
- Analyse and evaluate system performance using measurement techniques and analytical models to draw evidence-based conclusions.
- Interpret and model system performance using principles from queueing theory and operational analysis.
- Evaluate and justify the use of different performance analysis methodologies (e.g., USE method, workload characterization, latency analysis, static tuning, bottleneck analysis) for specific system scenarios.
- Optimize system performance by identifying bottlenecks and designing appropriate performance improvements.

Syllabus:

- Performance Analysis Methodologies
- Capacity Planning
- Queueing Theory Basics
- Markov Models
- Operational Analysis
- Observability Tools for Performance Analysis
- Resources Analysis (System)
- Monitoring and Visualization
- Applications Performance and Benchmarking

Grading policy: 100% CA:

- 20% - Test 1 (Tue, week 6)
- 10% - Paper review (due on Tue, week 11)
- 30% - Test 2 (Tue, week 12, in-person, open book)
- 5% - Presentation of the project or paper (Tue, week 13)
- 35% - Project (due on Tue, reading week)